FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale (short)
FR-001	Functional	The system shall allow authorized faculty to record new attendance and update previously recorded attendance for students in a selected subject and date.	High	When authorized faculty records attendance, the record is saved and appears in the student's attendance details. When faculty updates previously recorded attendance, the changes are saved and reflected in the attendance details.	Required for maintaining attendance records.
FR-002	Functional	The system shall calculate the attendance percentage for each student and subject.	Medium	For valid attendance data, the displayed percentage matches the calculated value based on classes attended and total classes.	Helps students track attendance
FR-003	Functional	The system shall allow faculty to enter and update marks for each student.	high	After marks are submitted, the updated marks are displayed in the student's academic record	Required for maintaining academic performance.

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FR-004	Functional	The system shall calculate semester GPA by converting student marks into grade points according to the institution's grading rules and using the corresponding subject credits.	High	When student marks are entered, the system converts them into grade points according to the institution's grading rules and calculates the semester GPA using the corresponding subject credits. The calculated GPA is displayed correctly for the selected semester.	Provides academic performance information

FR-005	Functional	The system shall allow students to view their attendance, marks, grades, GPA, and CGPA and select a semester to view the corresponding academic records.	Medium	When a student selects a semester, the system displays the corresponding attendance, marks, grades, and GPA. The system also displays the student's CGPA and ensures that only the student's own academic records are accessible.	Provides students access to their records.

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FR-006	Functional	The system shall allow users to log in and log out securely	High	Valid credentials allow access; invalid credentials are rejected.	Controls access to the system.
FR-007	Functional	The system shall allow authorized faculty to search for students using their student ID and view the corresponding student profile details.	Medium	When authorized faculty search for a student using the student ID, the system displays the matching student's profile. If no matching student is found, the system displays an appropriate message.	Helps faculty identify students.
FR-008	Functional	The system shall maintain subject details and corresponding credit values.	High	Each subject has a valid credit value stored in the system	Supports GPA calculations.
FR-009	Functional	Calculate CGPA using semester results and the corresponding subject credits, according to the institution's prescribed formula.	High	CGPA matches the institution's prescribed calculation formula.	Maintains cumulative academic results

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FR-010	Functional	The system shall allow faculty to filter attendance records by subject and date	Medium	Selected filters return only matching attendance records.	Makes attendance review easier.
FR-011	Functional	The system shall validate entered marks against the maximum permitted marks.	High	Marks within the permitted range are accepted; out-of-range marks are rejected.	Prevents invalid marks.
FR-012	Functional	The system shall generate a class-wise attendance report for faculty.	Medium	The report displays attendance details for students in the selected class or subject.	Helps faculty monitor attendance
FR-013	Functional	Notify students when their attendance percentage falls below the minimum attendance threshold specified by the institution.	Medium	A notification is generated when attendance falls below the configured threshold.	Alerts students to low attendance
FR-014	Functional	The system shall allow faculty to view a subject-wise summary of students' marks.	Medium	Faculty can see the marks of students enrolled in the selected subject.	Supports class performance analysis.

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FR-015	Functional	The system shall allow students to download their academic records as a report	Medium	The downloaded report contains the student's available academic records.	Provides a portable academic summary.
NFR001	Nonfunc.	The system shall display requested attendance and academic records within 2 seconds for at least 95% of requests under normal operating conditions.	Medium	At least 95% of requests display the requested records within 2 seconds under normal operating conditions	Ensures quick access to records.
NFR002	Nonfunc.	The system shall restrict access to academic records based on user roles. Students shall access only their own academic records, while faculty shall access records they are authorized to view or manage.	High	An unauthorized user attempting to access another student's record is denied access in every tested attempt.	Protects confidential student information.
NFR003	Nonfunc.	The system shall provide a user-friendly interface for students and faculty. At least 4 out of 5 test users shall be able to access attendance and GPA records without assistance.	Medium	At least 4 out of 5 test users can access attendance and GPA records without assistance	Makes the system easy to use.

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NFR004	Nonfunc	The system shall be available for at least 99% of the scheduled college operating hours.	High	The system shall be accessible for at least 99% of the scheduled operating hours during testing.	Ensures users can access records when required.
NFR005	Nonfunc	The system shall support backup and recovery of student records and allow a deleted test record to be successfully restored from a backup	Medium	A backup can be created and a deleted test record can be successfully restored from the backup	Prevents permanent loss of student data.
NFR006	Nonfunc	The system shall maintain the accuracy and consistency of student academic records and prevent unauthorized or invalid modifications.	High	Valid updates are saved correctly, and invalid modifications are rejected without corrupting existing records.	Ensures the reliability and accuracy of academic records.